MAS1004 Data & AI, 2026 Fall

Assignment 1: Short report

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My experience

For my categories I chose 4 Sanrio characters: Hello Kitty, Cinnamoroll, Pompompurin and Kuromi. The reason why I picked those 4 is that I am a big fan of Sanrio and I generally love cute things, so I thought that it would be nice to make this project a little bit cuter. And as they seem to be easily distinguishable, I assumed that training would go more smoothly. My prediction was that the color scheme and special shapes are the most obvious visual difference.

It was my first time working in VS Code, using GitHub and training model, so I was a little afraid that the assignment might be too difficult for me, as only took basic Python course before. However, even though it was complicated, it was manageable. Steps 2 to 5 were the most challenging for me because I had to ask Claude to break down and explain in an easy way how to do every single change. After I got used to using agents other steps went much faster. I was actually surprised how fast it was to train a model on my Macbook (each run took only 2-4 minutes). Making my cleaning rule and deleting junk took around 1 hour. The most enjoyable part was publishing my demo and seeing my hard work come to life. Testing and playing around with different input options (webcamera, upload image, drawing) was quite fun too. Also, while playing around, I noticed some patterns that significantly influenced the results, and those observations helped me a lot with Problem 7.

As for my training and test results, they were very high, but in reality model was not as accurate. I found out that the color of image is the main factor. For example, if picture is black, gray, dark in general or has a dark background, it almost guaranteed to be classified as Kuromi; anything yellow is Pompompurin; blue is Cinamonroll. I think, my model did not really learned well to distinguish based on the shape of the object. It can tell apart characters in grayscale only whe the picture is very clear (high contrast, plain background, object centered). Therefore, there is still room for improvement, maybe I should have found more diverse images for my training material.

After completing this project, I learned that even big tasks can be done, if we split them into small steps. Now, asI have become much more familiar with coding, I hope next assignment will go more smoothly. And now I know what possible difficulties and mistakes I can try to avoid in the future.



Figure 1: Original photo.



Figure 2: Grayscale photo.

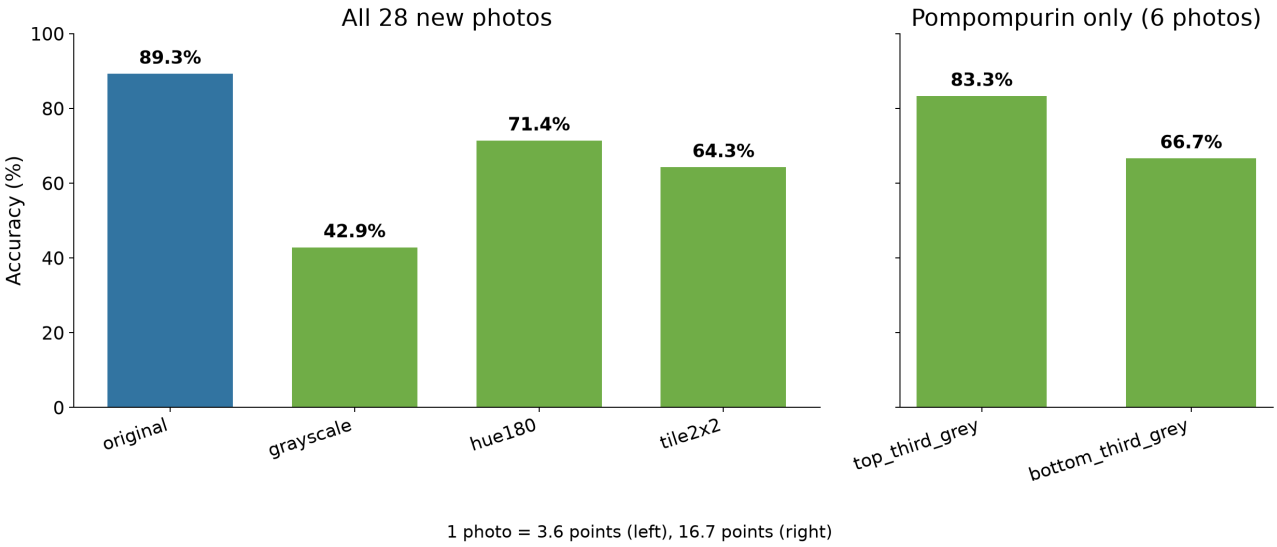


Figure 3: Accuracy after changing the pictures.